# GMAT

# Quantitative Reasoning

## Core

Advanced mathematical concepts and problem-solving techniques for the GMAT Quantitative Reasoning section.

This guide covers word problems (rate, work, mixture), statistics, sets, counting methods, probability, estimation, and sequences.

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**Educational Purpose:** This material is intended solely for educational purposes to help students prepare for standardized tests.

# 1. Rate, Work, and Mixture Problems

These word problem types appear frequently on the GMAT and follow consistent patterns.

## 1.1. Rate Problems

### 1.1.1. The Fundamental Formula

**Distance = Rate x Time**

Or equivalently:
- Rate = Distance / Time
- Time = Distance / Rate

**Example**: How many kilometers did a car travel in 4 hours at an average speed of 70 km/h?

Distance = Rate x Time = $70 \times 4 = 280$ km

### 1.1.2. Average Speed Problems

**Critical Concept**

Average speed = Total Distance / Total Time

**NOT** the average of the individual speeds!

**Example**: On a 600 km trip, a car went half the distance at 60 km/h and the other half at 100 km/h. What was the average speed?

**Step 1**: Find time for each half

First 300 km: $\frac{300}{60} = 5$ hours

Second 300 km: $\frac{300}{100} = 3$ hours

**Step 2**: Calculate average speed

Average speed = $\frac{600}{5+3} = \frac{600}{8} = 75$ km/h

Note: This is NOT $\frac{60+100}{2} = 80$ km/h!

### 1.1.3. Relative Speed Problems

When two objects move toward or away from each other:

**Relative Speed Rules**

- **Moving toward each other**: Add the speeds
- **Moving away from each other**: Add the speeds
- **Moving in same direction**: Subtract the speeds

**Example**: Two trains 300 km apart travel toward each other at 60 km/h and 40 km/h. When will they meet?

Combined speed = $60 + 40 = 100$ km/h

Time to meet = $\frac{300}{100} = 3$ hours

## 1.2. Work Problems

### 1.2.1. The Work Formula

**Basic Work Formula**

If person A takes $r$ hours to complete a job, and person B takes $s$ hours to complete the same job, then working together they take $h$ hours, where:

$$\frac{1}{r} + \frac{1}{s} = \frac{1}{h}$$

Think of it as: (A's rate) + (B's rate) = (Combined rate)

**Example**: Machine A takes 4 hours to make 1,000 bolts. Machine B takes 5 hours. How long do both machines working together take?

$$\frac{1}{4} + \frac{1}{5} = \frac{1}{h}$$

$$\frac{5}{20} + \frac{4}{20} = \frac{1}{h}$$

$$\frac{9}{20} = \frac{1}{h}$$

$$h = \frac{20}{9} = 2\frac{2}{9}$$

hours

## 1.3. Mixture Problems

### 1.3.1. Basic Mixture Problems

**Example**: If 6 kg of nuts costing $1.20/kg are mixed with 2 kg of nuts costing $1.60/kg, what is the cost per kg of the mixture?

Total cost = $6(1.20) + 2(1.60) = 7.20 + 3.20 = 10.40$

Total weight = $6 + 2 = 8$ kg

Cost per kg = $\frac{10.40}{8} = 1.30$/kg

### 1.3.2. Mixture Problems with Percents

**Example**: How many liters of a 15% salt solution must be added to 5 liters of an 8% salt solution to make a 10% salt solution?

Let $n$ = liters of 15% solution needed

**Set up equation**: Salt in 15% + Salt in 8% = Salt in mixture

$$0.15n + 0.08(5) = 0.10(n + 5)$$

$$0.15n + 0.40 = 0.10n + 0.50$$

$$0.05n = 0.10$$

$$n = 2$$

liters

**Mixture Problem Strategy**

1. Identify what quantity is being “preserved” (salt, cost, etc.)
2. Write an equation: Amount in Part 1 + Amount in Part 2 = Amount in Mixture
3. Solve for the unknown

# 2. Statistics

## 2.1. Measures of Central Tendency

### 2.1.1. Mean (Average)

**Mean Formula**

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

**Example**: Find the mean of 6, 4, 7, 10, 4

Mean = $\frac{6+4+7+10+4}{5} = \frac{31}{5} = 6.2$

### 2.1.2. Median

The **median** is the middle value when data is ordered.

- **Odd number of values**: The middle number
- **Even number of values**: Average of the two middle numbers

**Example 1** (odd): Find the median of 6, 4, 7, 10, 4

Order: 4, 4, 6, 7, 10

Median = 6 (middle value)

**Example 2** (even): Find the median of 4, 6, 6, 8, 9, 12

Median = $\frac{6+8}{2} = 7$

### 2.1.3. Mode

The **mode** is the value that occurs most frequently.

**Example**: The mode of 1, 3, 6, 4, 3, 5 is **3** (occurs twice)

## 2.2. Measures of Dispersion

### 2.2.1. Range

**Range = Maximum value - Minimum value**

**Example**: Range of 11, 10, 5, 13, 21 is $21 - 5 = 16$

### 2.2.2. Standard Deviation

**Standard deviation** measures how spread out values are from the mean.

**Steps to Calculate Standard Deviation**

1. Find the mean
2. Find each value's difference from the mean
3. Square each difference
4. Find the average of the squared differences
5. Take the square root

**Example**: Find the standard deviation of 0, 7, 8, 10, 10 (mean = 7)

| $x$ | $x - 7$ | $(x-7)^2$ |
|---|---|---|
| 0 | $-7$ | 49 |
| 7 | 0 | 0 |
| 8 | 1 | 1 |
| 10 | 3 | 9 |
| 10 | 3 | 9 |
| **Total** | | **68** |

Standard deviation = $\sqrt{\frac{68}{5}} \approx 3.7$

# 3. Sets

## 3.1. Basic Definitions

A **set** is a collection of elements. Sets are written with braces: $\{-5, 0, 1\}$

**Notation**

- $|S|$ = number of elements in set $S$ (cardinality)
- $A \cup B$ = union (elements in $A$ or $B$ or both)
- $A \cap B$ = intersection (elements in both $A$ and $B$)

**Examples**:
- $\{3, 4\} \cup \{4, 5, 6\} = \{3, 4, 5, 6\}$
- $\{3, 4\} \cap \{4, 5, 6\} = \{4\}$

## 3.2. The General Addition Rule

**General Addition Rule for Sets**

$$|A \cup B| = |A| + |B| - |A \cap B|$$

If $A$ and $B$ are disjoint: $|A \cup B| = |A| + |B|$

**Example**: Each of 25 students takes history, mathematics, or both. If 20 take history and 18 take mathematics, how many take both?

Let $n$ = students taking both

Using the addition rule:

$$25 = 20 + 18 - n$$

$$n = 38 - 25 = 13$$

**13 students take both subjects.**

# 4. Counting Methods

## 4.1. The Multiplication Principle

**Basic Multiplication Principle**

If one choice can be made in $m$ ways and another independent choice in $n$ ways, then both choices can be made in $m \times n$ ways.

**Example**: A meal must include 1 entree (5 options) and 1 dessert (3 options).

Total meals = $5 \times 3 = 15$

## 4.2. Factorials

**Factorial Definition**

For integer $n > 1$:

$$n! = n \times (n-1) \times (n-2) \times ... \times 2 \times 1$$

Special cases: $0! = 1! = 1$

**Examples**:
- $3! = 3 \times 2 \times 1 = 6$
- $4! = 4 \times 3 \times 2 \times 1 = 24$
- $5! = 120$

## 4.3. Permutations and Combinations

A **permutation** is an ordered arrangement of elements.

A **combination** is a selection without regard to order.

**Combination Formula**

The number of ways to choose $k$ objects from $n$ objects:

$$C(n,k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

**Example**: Choose 2 letters from $\{A, B, C, D, E\}$

$C(5,2) = \frac{5!}{2! \times 3!} = \frac{120}{2 \times 6} = 10$

# 5. Probability

## 5.1. Basic Definitions

**Probability** measures the likelihood of an event occurring.

**Probability Range**

For any event $E$:

$$0 \leq P(E) \leq 1$$

- $P(E) = 0$: Event is impossible
- $P(E) = 1$: Event is certain

## 5.2. Equally Likely Outcomes

When all outcomes are equally likely:

$$P(E) = \frac{\text{Number of outcomes in } \ E}{\text{Total number of possible outcomes}}$$

**Example**: A fair 6-sided die is rolled. What is the probability of an odd number?

Favorable outcomes: $\{1, 3, 5\}$ (3 outcomes)

Total outcomes: 6

$P(\text{odd}) = \frac{3}{6} = \frac{1}{2}$

## 5.3. Compound Events

**Probability Rules**

- **Complement**: $P(\text{not } \ E) = 1 - P(E)$
- **Addition Rule**: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- **If mutually exclusive**: $P(A \text{ or } B) = P(A) + P(B)$

### 5.4. Independent Events

Two events are **independent** if neither affects the other's probability.

**Multiplication Rule for Independent Events**

$$P(A \text{ and } B) = P(A) \times P(B)$$

### 5.5. Dependent Events

Events are **dependent** if one affects the other's probability.

**General Multiplication Rule**

$$P(A \text{ and } B) = P(A \mid B) \times P(B)$$

where $P(A \mid B)$ is the probability of $A$ given that $B$ occurs.

## 6. Estimation

### 6.1. Rounding

**Rounding** simplifies complex calculations by using nearby numbers with fewer digits.

**Examples**:
- Round 7651.4 to nearest hundred: 7700
- Round 0.43248 to nearest thousandth: 0.432

**When to Use Estimation**

- Answer choices are spread far apart
- Exact calculation is too time-consuming
- The question asks for an approximate value
- You can eliminate obviously wrong answers

# 7. Sequences and Series

## 7.1. Sequences

A **sequence** is a function whose domain is positive integers.

Notation: $a_n$ represents the $n$th term.

**Arithmetic sequence**: Constant difference between terms

$a_n = a_1 + (n-1)d$ where $d$ is the common difference

**Geometric sequence**: Constant ratio between terms

$a_n = a_1 \times r^{n-1}$ where $r$ is the common ratio

## 7.2. Series

A **series** is the sum of a sequence's terms.

### 7.2.1. Sum of Arithmetic Series

$$S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n-1)d)$$

**Example**: Sum of first 100 positive integers

$S_{100} = \frac{100}{2}(1 + 100) = 50 \times 101 = 5050$

# 8. Problem-Solving Techniques

## 8.1. Backsolving

Test answer choices in the original problem.

**When to Use Backsolving**

- Numeric answer choices
- Problem would require complex algebra
- Start with middle value (choice C or B)

**Example**: If $x^2 - 5x = 14$, find $x$.

Choices: (A) −7 (B) −2 (C) 2 (D) 7 (E) 14

Test (C): $2^2 - 5(2) = 4 - 10 = -6 \neq 14$ ×

Test (D): $7^2 - 5(7) = 49 - 35 = 14$ ✓

Answer: (D)

## 8.2. Number Picking

Choose easy values for variables.

**When to Use Number Picking**

- Variables in answer choices
- Percent or ratio problems without specific numbers
- Abstract relationships

**Example**: If $x$ is increased by 25%, the result is what percent of $2x$?

Pick $x = 100$

Increased: $100 + 25 = 125$

Percent of $2x$: $\frac{125}{200} = 62.5\%$

## 8.3. Process of Elimination

Eliminate clearly wrong answers to improve odds.

Even eliminating one or two choices significantly improves your probability of guessing correctly if you need to move on.

## 9. Summary

This Core guide has covered:

**Word Problems**:
- Rate problems and average speed
- Work problems with multiple workers/machines
- Mixture problems with percents

**Statistics**:
- Mean, median, mode
- Range and standard deviation

**Sets and Counting**:
- Set operations and Venn diagrams
- Multiplication principle
- Permutations and combinations

**Probability**:
- Basic probability and equally likely outcomes
- Independent and dependent events
- Compound probability rules

**Additional Topics**:
- Estimation and rounding
- Sequences and series
- Problem-solving techniques

Master these Core concepts before moving on to the Excellence guide, which covers advanced optimization strategies and time-saving techniques.